

Numerical Analysis of the Airy Equation in the WKB Approximation

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Abstract

This report focuses on the Airy equation, an important ordinary differential equation in the WKB approximation. We will compare different numerical methods and investigate how to address the stiffness of the equation on the positive interval.

Keywords: WKB Approximation, Airy Equation, Numerical Analysis

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1 Background

1.1 WKB Approximation

When studying real systems, it is very complicated to solve the wave function using quantum mechanics. For this reason, we need to use approximate methods to solve the Schrödinger equation. The WKB approximation is a semiclassical approximation method suitable for systems where the potential energy changes slowly relative to the wavelength of the particle.

According to [1] & [3], the derivation of the WKB approximation can be obtained by various methods. We give the final result here:

$$\varphi(x) = \frac{C_{\pm}}{\sqrt{k(x)}} e^{\pm i \int^x k(t) dt} \quad (1)$$

$$k(x) = \begin{cases} \sqrt{\frac{2m}{\hbar^2}(E - V(x))} & \text{where } E > V(x), \\ -i\sqrt{\frac{2m}{\hbar^2}(E - V(x))} & \text{where } E < V(x). \end{cases} \quad (2)$$

1.2 Airy Equation

1.2.1 Airy Equation in WKB Approximation

It's easy to use the WKB approximation to solve the classical and non-classical regions, but what we really care about is the situation at the turning point. In this section, we will focus on the connection formulas for turning points and introduce the Airy equation.

At the turning point, where $E = V$, the WKB approximation itself breaks down. Now, what we need to do is splice the two WKB solutions together, using a patching wave function that straddles the turning point.

We assume the turning point is the origin. Since we only need the patching wave function φ in the neighborhood of the origin, we'll approximate the potential by a straight line:

$$V(x) \approx E + V'(0)x \quad (3)$$

and then solve the Schrödinger equation for this linearized V :

$$\begin{aligned} -\frac{\hbar^2}{2m} \frac{d^2\varphi}{dx^2} + (E + V'(0)x)\varphi &= E\varphi \\ \Leftrightarrow \frac{d^2\varphi}{dx^2} &= \alpha^3 x \varphi, \quad \text{where } \alpha \equiv \sqrt[3]{\frac{2m}{\hbar^2} V'(0)} \end{aligned} \quad (4)$$

It is easy to find that this equation can be simplified to:

$$\frac{d^2\varphi}{dz^2} = z\varphi, \quad \text{where } z = \alpha x \quad (5)$$

This is the **Airy equation**, and its solutions are **Airy functions**. Since the Airy equation is a second-order differential equation, there are two linearly independent Airy functions. The general solution can be written as:

$$\varphi(x) = a\text{Ai}(z) + b\text{Bi}(z) \quad (6)$$

1.2.2 Analytical Solution of Airy Equation

From a mathematical perspective, the Airy equation, as a simple second-order ordinary differential equation, has the potential to obtain its analytical solution. In fact, after trying, we found that the analytical solution corresponding to the Airy equation can be obtained through two methods: power series expansion and Fourier transform. In this section, we will demonstrate the use of Fourier transform to solve two sets of linearly independent solutions $\text{Ai}(x)$ and $\text{Bi}(x)$:

$$\text{Ai}(x) = \frac{1}{\pi} \int_0^{\infty} \cos\left(\frac{t^3}{3} + tx\right) dt \quad (7)$$

$$\text{Bi}(x) = \frac{1}{\pi} \int_0^{\infty} e^{(-\frac{t^3}{3} + xt)} + \sin\left(\frac{t^3}{3} + xt\right) dt \quad (8)$$

Assuming this equation has a square integrable solution $\varphi(x)$. The Fourier transformation of $\varphi(x)$ is defined as:

$$\Phi(t) = \int_{-\infty}^{+\infty} \varphi(x) e^{-itx} dx \quad (9)$$

So using the properties of Fourier transform, it is obvious that we have:

$$\int_{-\infty}^{+\infty} \varphi''(x) e^{-itx} dx = -t^2 \Phi(t) \quad (10)$$

and,

$$\int_{-\infty}^{+\infty} x \varphi(x) e^{-itx} dx = i \Phi'(t) \quad (11)$$

Apply Fourier transformation to both sides of Airy Equation (Eq.(5)), we get:

$$t^2 \Phi + i \Phi' = 0 \quad (12)$$

which is a first order ODE of Φ with respect to t . Do separation of variables and integrate on both sides, we obtain its solution:

$$\Phi(t) = A e^{i \frac{t^3}{3}} \quad (13)$$

where A is an arbitrary constant.

Apply the inverse Fourier transform to our solution, we find:

$$\varphi(x) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \Phi(t) e^{itx} dt = \frac{1}{2\pi} \int_{-\infty}^{+\infty} A e^{i(\frac{t^3}{3} + tx)} dt \quad (14)$$

Further simplify our integral, Eq.(14) equals:

$$\begin{aligned} \varphi(x) &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} A e^{i(\frac{t^3}{3} + tx)} dt \\ &= \frac{A}{2\pi} \int_{-\infty}^{+\infty} \left[\cos\left(\frac{t^3}{3} + tx\right) + i \sin\left(\frac{t^3}{3} + tx\right) \right] dt \\ &= \frac{A}{2\pi} \int_{-\infty}^{+\infty} \cos\left(\frac{t^3}{3} + tx\right) dt \\ &= \frac{A}{\pi} \int_0^{+\infty} \cos\left(\frac{t^3}{3} + tx\right) dt \end{aligned} \quad (15)$$

Here, in the second equal sign, we used Euler's formula, while for the third and fourth equal sign, we use the asymmetric of odd equation ($\sin(x)$) and the symmetric of even equation ($\cos(x)$) respectively.

After obtaining the first analytical solution of the Airy equation, we can use the scaling transformation method combined with the contour integral to obtain the second linearly independent analytical solution $\text{Bi}(x)$. The explicit expression of the second solution is shown in Eq.(8).

1.2.3 Asymptotic Forms

The behavior of the Airy function at positive infinity is also an important part of studying the connection formula of the WKB approximation. In [4], we use the saddle point approximation to solve this problem, and the final result is:

$$\left. \begin{aligned} \text{Ai}(z) &\simeq \frac{1}{2\sqrt{\pi}\sqrt[4]{z}} \exp\left(-\frac{2}{3}z^{3/2}\right) \\ \text{Bi}(z) &\simeq \frac{1}{\sqrt{\pi}\sqrt[4]{z}} \exp\left(\frac{2}{3}z^{3/2}\right) \end{aligned} \right\} \quad z \gg 0 \quad (16)$$

On the positive interval $x \gg 0$, the asymptotic behaviors of the two solutions are very different. $\text{Ai}(x)$ decays exponentially, while $\text{Bi}(x)$ grows exponentially. These contrasting asymptotic behaviors cause stiffness in the numerical solution, as will be discussed in the next section.

2 Methodology

2.1 Numerical Solution of Ordinary Differential Equation

In the above analysis, we have given the analytical solution of the Airy function through Fourier transformation. However, since they can only be written in the form of integrals or series, we cannot intuitively understand the behavior of these functions. In this section, we will use numerical methods to solve the Airy equation, depict the image of the Airy function and analyze the relative errors. In this report, we focus on $\text{Ai}(x)$.

2.1.1 Initial Value Problem

Before we start, we need to give an initial condition. The easiest value to calculate is the function at $x = 0$, where the integral degenerates to:

$$\text{Ai}(0) = \frac{1}{\pi} \int_0^\infty \cos\left(\frac{t^3}{3}\right) dt = \frac{1}{\pi} \int_0^\infty \cos(u) u^{-2/3} du \quad (17)$$

This integral can be solved by Laplace transform, and finally we get:

$$\int_0^\infty \cos(u) u^{v-1} du = \Gamma(v) \cos\left(\frac{\pi v}{2}\right), \quad v \in (0, 1) \quad (18)$$

and then:

$$\text{Ai}(0) = \frac{1}{\pi} \Gamma\left(\frac{1}{3}\right) \cos\left(\frac{\pi}{6}\right) = \frac{1}{3^{2/3} \Gamma(2/3)} \quad (19)$$

The second initial condition is the derivative of the Airy function. We can directly differentiate the integral to get:

$$\text{Ai}'(x) = -\frac{1}{\pi} \int_0^\infty t \sin\left(\frac{t^3}{3} + xt\right) dt \quad (20)$$

Similarly, we take the same approach to the derivative value at $x = 0$:

$$\text{Ai}'(0) = -\frac{1}{\pi} \int_0^\infty t \sin\left(\frac{t^3}{3}\right) dt = -\frac{\sqrt[3]{3}}{\pi} \int_0^\infty \sin(u) u^{-1/3} du \quad (21)$$

This integral can also be solved by Laplace transform, and the final result is:

$$\int_0^\infty \sin(u) u^{v-1} du = \Gamma(v) \sin\left(\frac{\pi v}{2}\right), \quad v \in (0, 1) \quad (22)$$

$$\text{Ai}'(0) = -\frac{\sqrt[3]{3}}{\pi} \Gamma\left(\frac{2}{3}\right) \sin\left(\frac{\pi}{3}\right) = -\frac{1}{3^{1/3} \Gamma(1/3)} \quad (23)$$

2.1.2 Explicit Method

After determining the initial conditions for the first linear independent solution $\text{Ai}(x)$, namely the function value and derivative at the origin, we can consider using the three explicit numerical methods mentioned in [2] to solve the Airy equation.

In this section, we used the Euler method, midpoint method (also called RK2), and fourth-order Runge-Kutta method (RK4) to solve the Airy equation and plotted their corresponding relative error graphs, as shown in Figure 1.

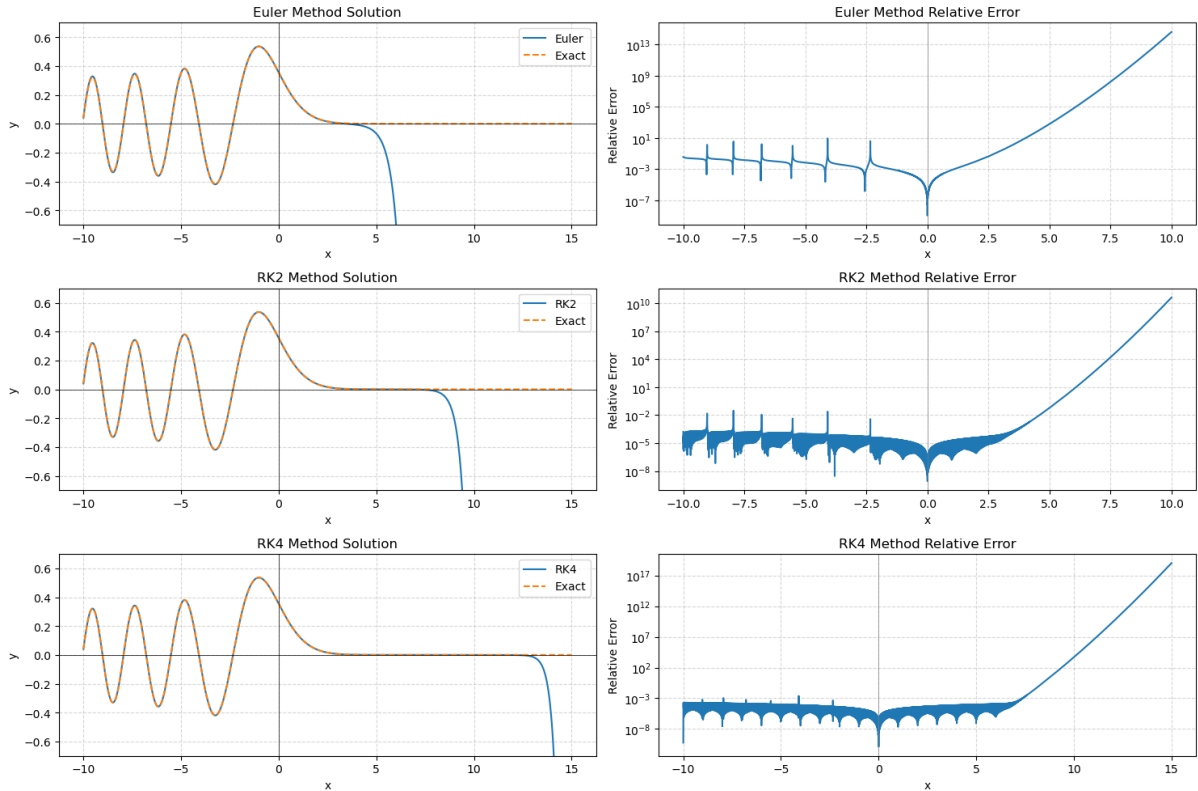


Figure 1: Explicit Method and Their Relative Error

Observing the output results of these three methods, we can see that as the accuracy of our algorithm improves, the stability of the output results does gradually improve. However, these methods all face the problem of divergence caused by error accumulation in the positive half axis of the x-axis, deviating from the theoretical true value.

- Explanation for the divergent behavior of solutions on the positive x-axis:

From section 1.2.3, we learnt that Airy equation have two linearly independent solutions $Ai(x)$ and $Bi(x)$ which have totally different asymptotic behavior: the former one decrease exponentially while the later one increase exponentially. In the area of $x \gg 1$, although the true solution should be $Ai(x)$ since the initial condition doesn't have $Bi(x)$, Numerical errors or rounding errors may also trigger it. Once $Bi(x)$ is triggered, it will lead to a grand increasing, which make our equation stiff.

2.1.3 Implicit Method

After realizing the stiffness of Airy equation, we came up with a idea of modifying these explicit method into implicit form to obtain a more stable solution.

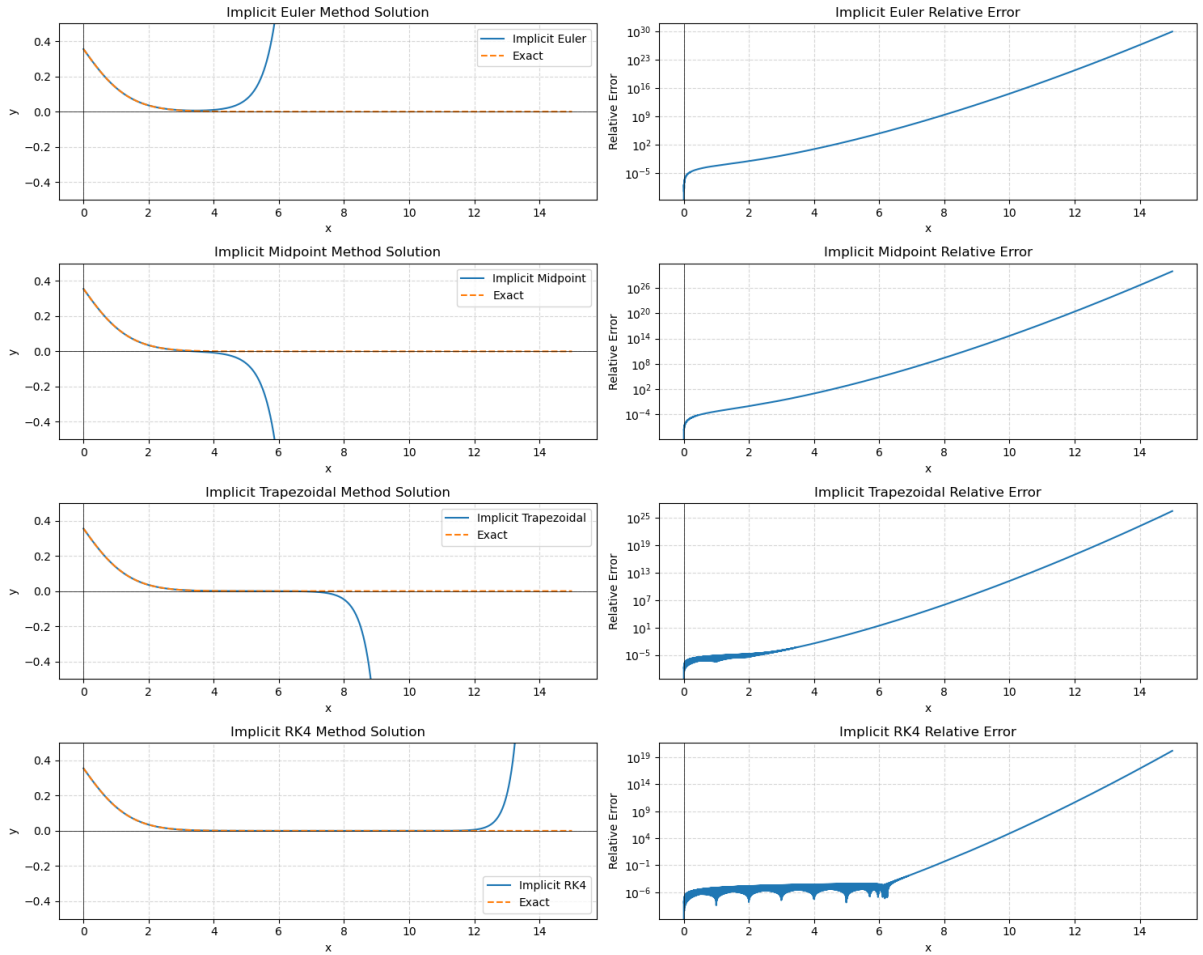


Figure 2: Implicit Method and Their Relative Error

We used four different implicit methods to deal with Airy function. (See Figure 2) These implicit methods do show some improvement in stability compared to explicit

methods. Also, as the accuracy of our algorithm increases, the region of stability of the solution also expands. However, the Airy equation is stiff enough on the positive axis that we cannot solve the divergence problem by simply rewriting the explicit form into the implicit form.

Since both explicit and implicit methods cannot solve the stiffness of the Airy equation, how can we keep the solution stable? We found that the divergence of the numerical solution occurs at $x \gg 0$, so if we add new restrictions far enough away, can we solve the stiffness of the Airy equation? To this end, we proposed two new methods: the shooting method and the backward method.

2.1.4 Shooting Method

If we add new restrictions far enough away, we can treat this problem as a boundary problem. For example, we set the boundary value of Airy equation at $x = 0$ and $x = 15$. We already have an analytical solution for the former, and we can find the latter through asymptotic form. When solving this type of second-order boundary problem, we can use the shooting method.

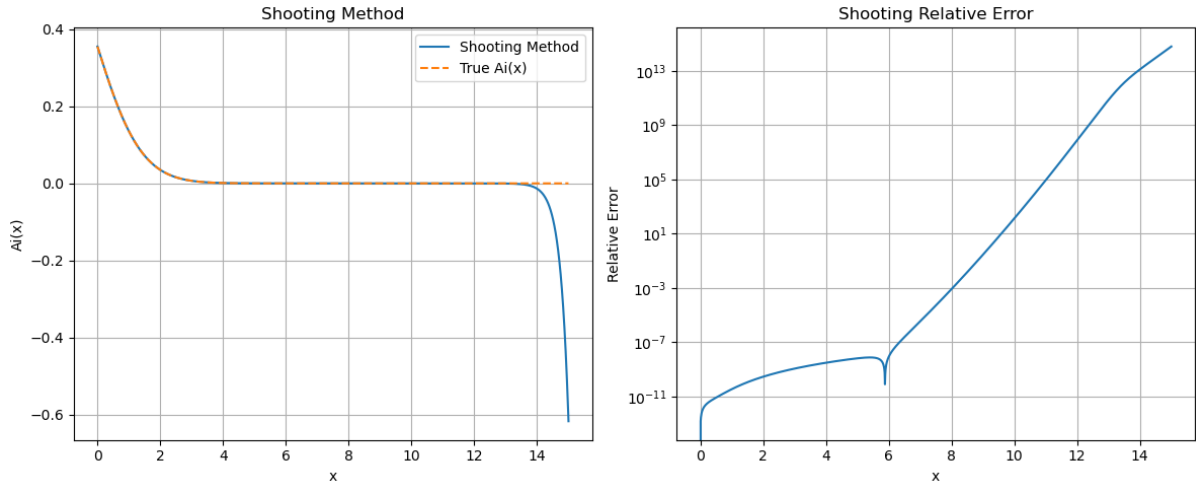


Figure 3: Shooting Method and Its Relative Error

The far-point boundary condition can be obtained by Eq.(16), where we take the approximation when $x = 15$ as the new boundary condition. The final result is shown in Figure 4, where we find that the shooting method also seems unable to solve the stiffness of the system.

2.1.5 Backward Method

The rigidity of solving the Airy equation is essentially to avoid the contamination of $Ai(x)$ by $Bi(x)$. The shooting method tried to eliminate the influence of $Bi(x)$ by boundary restrictions, but failed. It is like we adjust the angle of throwing the ball to hit the target we set, but found that no matter how we throw it, once it enters a larger range, there is always a strong field that distorts its trajectory.

This distortion comes from the influence of the initial error, and even though we have given an analytical solution, it cannot be avoided. Because of the exponential growth of $Bi(x)$, even a slight deviation is enough to trigger a result that is far different from the

exponentially decaying $\text{Ai}(x)$. For example, even if the deviation of my initial value only makes my numerical solution system contain $\text{Bi}(x)$ component with $b = 10^{-16}$. But when $x = 15$, $\text{Bi}(x) \sim 10^{16}$, making the numerical solution about 1 more than the true value. At this time, $\text{Ai}(x) \sim 10^{-18}$, and the deviation causes the extra value to be 10^{18} times that of $\text{Ai}(x)$. This also shows that the Airy equation has strong sensitivity and stiffness in the positive interval, and starting from the origin may not be a good solution.

Now the path we can choose is obvious. We need to start from $x \gg 0$ and solve the equation backward. Because at the far point x_k , $\text{Bi}(x_k) \gg \text{Ai}(x_k)$, if we can give a good enough restriction to make $b\text{Bi}(x_k) \ll a\text{Ai}(x_k)$, it means that our initial restriction is still valid when $x < x_k$. Then $\text{Bi}(x)$ will no longer be able to contaminate our numerical solution.

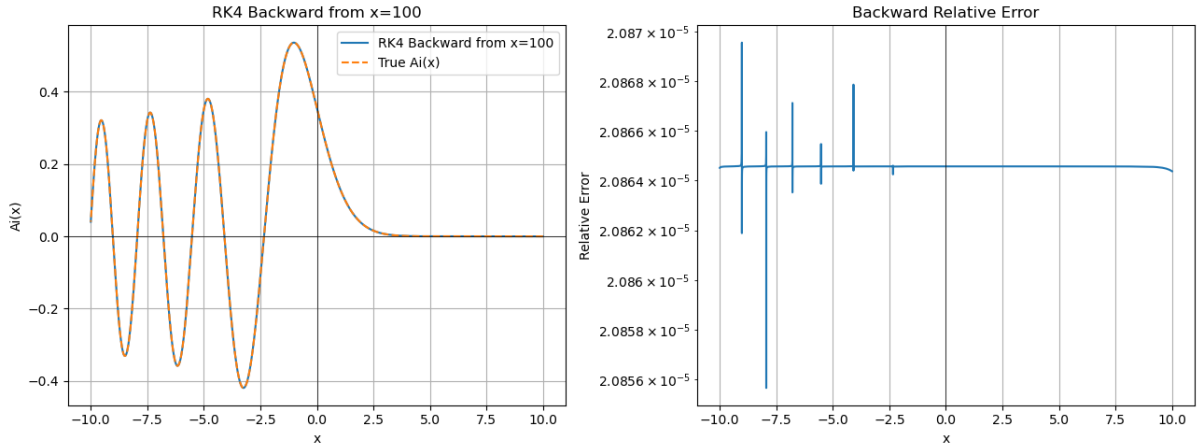


Figure 4: Backward Method and Its Relative Error

We set $x = 100$ as the far point, give the function value and derivative value by approximation, and start the backward solution. There is no doubt that this is a good enough point, at this time $\text{Bi}(x) \sim 10^{290}$. If the restrictions we give here can eliminate the contamination of $\text{Bi}(x)$, then we can naturally ensure the stability of $\text{Ai}(x)$ on the entire interval. The final result is shown in Figure 4. The backward method not only solves the stiffness of the positive interval, but also conforms to the oscillation of the system in the negative interval.

2.2 Numerical Integration

In section 1.2.2, we have derived the analytical solution of Airy equation $\text{Ai}(x)$ and $\text{Bi}(x)$, also are called “Airy function”. We can use numerical methods to calculate the integral expression of the Airy function at different x values to plot the corresponding function graph. To deal with this integral numerically, we should first map the integration interval from $(0, \infty)$ to finite area, say $(0, 1)$. We realize this mapping by following change of variables:

$$x = \frac{t}{1-t} \quad (24)$$

then,

$$dx = \left(\frac{u}{1-u}\right)' du = \frac{1}{(1-u)^2} du \quad (25)$$

Therefore, the integral Eq.(7) becomes:

$$\text{Ai}(x) = \frac{1}{\pi} \int_0^1 \cos\left[\frac{u/(1-u)^2)^3}{3} + x \frac{u}{1-u}\right] / (1-u)^2 du \quad (26)$$

In the following part, we will discuss the numerical integration of this transformed expression.

2.2.1 Singularity and Determination of Upper Limit

When u approaches 1, the denominator of the dependent variable after we replace the variable will approach 0, causing the function to diverge, as shown in Figure .

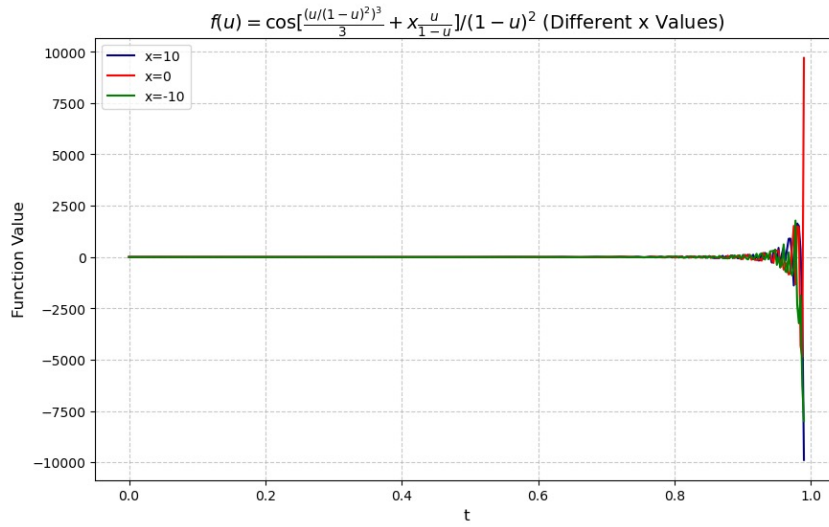


Figure 5: The Plot of Transformed Function $f(u)$

Near the singularity of $u = 1$, the function oscillates violently, causing instability in the integration result (as shown in Figure 6).

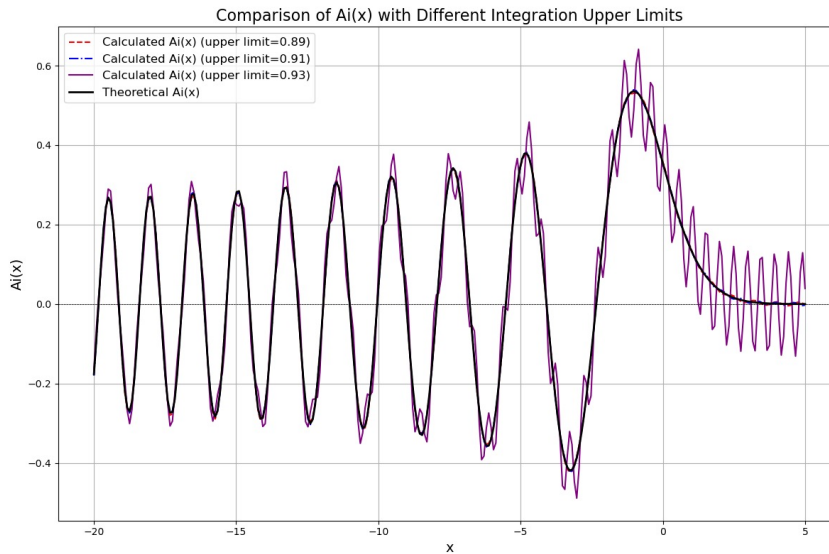


Figure 6: Function Plotted with Different Upper Limits

To avoid this singularity while ensuring sufficient information about the integrand function is captured, we have chosen $u = 0.91$ as our upper limit for integration based on the results shown in the Figure 5 & 6.

2.2.2 Analysis of Different Integration Methods Relative Error

After setting the upper limit of integration $u = 0.91$ and a uniform number of integration intervals $n = 5000$, we used three numerical integration methods, namely the rectangle rule, the trapezoidal rule, and the Simpson's rule, to calculate the Airy function and plotted the relative error between the calculated result and the theoretical value, as shown in Figure 7.

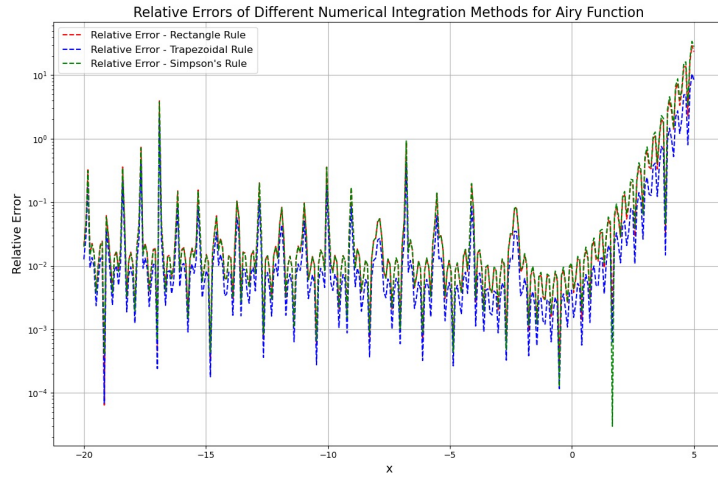


Figure 7: Relative Error with Different Integration Methods

From the above figure, we conclude that the trapezoidal method is the most accurate among the three methods when other parameters remain unchanged. The Simpson's method, which is usually more accurate, is not as good as the basic rectangle rule and trapezoidal rule in this integral.

- Explanation for relative error behavior:

The anomalous relative error behavior is primarily attributed to the poor behavior of the integrand. Generally, Simpson's rule, Gaussian quadrature, and other methods offer higher precision. This conclusion, however, is premised on the integrand having good properties, such as low oscillation magnitude and continuous second derivatives. Conversely, the integrand in Eq.(26) is highly oscillatory. This frequent oscillation leads to a scenario where methods typically known for higher accuracy under perform compared to basic quadrature methods.

2.2.3 Final Output

Based on the above analysis, we finally chose the trapezoidal rule, with upper limit $u = 0.91$ & interval number $n = 5000$ to complete the characterization of this integral expression. The result is shown in the Figure below.

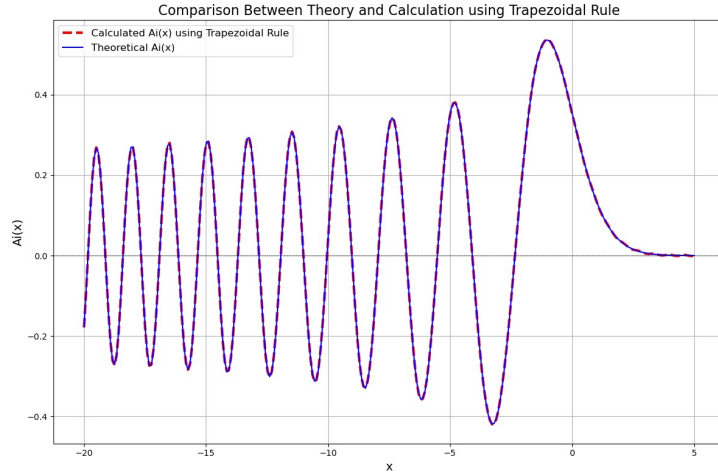


Figure 8: Final Integral Output with Trapezoidal Rule

3 Conclusion

In this report, we conducted a comprehensive numerical analysis of the Airy equation, which emerges near turning points in the WKB approximation. We used two methods to complete the discussion of it, namely numerical solution of differential equations and integrating the analytical solution.

For the part of solving differential equation, although explicit and implicit methods were implemented, they failed to overcome the stiffness of the equation in the positive domain due to the exponentially growing $\text{Bi}(x)$ component. We further explored the shooting method by imposing boundary conditions at distant points, but it still failed to suppress the instability caused by numerical contamination of $\text{Bi}(x)$, especially when integrating from the origin. In contrast, the backward integration method, initialized from a large positive value where $\text{Bi}(x)$ dominates, proved to be effective. By enforcing strict asymptotic constraints, it successfully isolated the $\text{Ai}(x)$ component and maintained stability across the entire domain. This underscores the necessity of directional integration and asymptotic control in solving stiff differential equations arising from semiclassical approximations.

As for numerical integration part, we first do change of variable to map the integration domain from infinite to a finite interval. Secondly, we found its singularity by analyzing the behavior of the integrand. It was found that near the singularity $u = 1$, the function value would diverge and oscillate violently, causing the integral result to lose stability. We solved this difficulty by properly adjusting the upper limit of the integral. Then, we compared the relative errors of the three numerical integration methods with the true value, found that the error of the trapezoidal rule was the smallest, and analyzed the reasons behind it. Finally, we used the trapezoidal method for numerical integration and output the final plot.

4 Further Discussion

The Airy equation serves as a prototypical model for studying turning point behavior in the WKB approximation, but the challenges encountered here are representative of a broader class of stiff differential equations in quantum mechanics and wave propagation.

The severe stiffness observed on the positive real axis reflects the inherent sensitivity of mixed-exponential solutions, a phenomenon that becomes more pronounced in multi-turning-point systems or when matching across classically forbidden regions.

Although backward integration successfully isolates the desired $\text{Ai}(x)$ solution in our study, this approach relies heavily on the availability of precise asymptotic expressions, which may not be feasible for more complex potentials or coupled systems. In such cases, adaptive step-size methods, spectral approaches, or basis transformation techniques (e.g., removing the dominant exponential growth analytically) could offer more generalizable alternatives.

Moreover, the failure of the shooting method highlights a key limitation of boundary value solvers in stiff contexts: sensitivity to initial slope guesses and round-off amplification. Exploring stabilization techniques, such as orthogonalization or projecting out unstable modes during integration, may improve robustness.

Future work could also explore extending these techniques to higher-order WKB corrections or semiclassical approximations in multidimensional systems, where the local Airy-type behavior generalizes to more intricate special functions and phase-space structures. These generalizations are essential for bridging numerical methods with modern applications in quantum tunneling, optics, and even cosmology.

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